

Designing of various schemes to Control a Dual active bridge converter for EV charging applications

Abstract: Dual active bridge converters are widely used in Electric vehicle charging because of their efficiency, bidirectional power transfer capability and galvanic isolation. There are two bridges each consisting of 4 active switches. One bridge act as inverter and other acts as a rectifier. Both bridges generate AC like square waves actively. Power transfer depends on phase difference between two bridges. For up to a certain limit larger the phase difference larger is the power transferred and vice versa.so we need to control phase angle to control power transfer.

There are two basic topologies possible Open loop and closed loop configuration. In open loop configuration there is no feedback from output to input so we can't control power transfer once parameters are initialised. But in closed loop there is feedback from output back to the input so power control is possible. However, because the converter switches on and off at an extremely high speed, the main coil—called the inductor—frequently creates unstable electrical spikes and messy current ripples whenever the power flows or changes. To solve this, the project presents a complete design and comparison of three different control setups. These setups are tested on a high-speed, 100 kilohertz converters using a simulation that updates every single microsecond. The three configurations compared are a traditional dual-loop Proportional-Integral (PI) controller, a basic Artificial Neural Network (ANN) helper structure, and an advanced Recurrent Neural Network (RNN) Long Short-Term Memory (LSTM) architecture.

Introduction: The demand for Electric vehicles is increasing very fast all over the world due to heavy pollution caused by combustion engine vehicles.so the demand for charging infrastructure and fast charging has emerged [1]. Among various DC-DC converters the Dual active bridge converter became the preferred choice for automotive applications. It is preferred for its bidirectional power transfer capability, efficiency and galvanic isolation. With this converter energy transfer from grid to vehicle is possible and vice versa [2].

Literature background:

The requirement for green transportation charging technology has accelerated significantly because of rapid global adoption of Electric vehicles [3]. Traditional charging methods depend heavily on linear control mathematical models such as PI controllers [4]. But classical linear controllers exhibit major design problems when we apply them to nonlinear bidirectional converters like Dual active bridge converter [5]. Research shows that linear techniques cannot handle systematic difficult situations like

uncertainties, variations in the state of charge of lithium batteries, and fast oscillations in output voltage and current [6].

To overcome control limitations recent work has shifted towards using AI and computational intelligence to make these controllers assist in a better way. Optimisation approach like Particle swarm optimization have been used to enhance energy management systems and automation of optimal parameter selection [7].

The focus has been shifted towards integrating AI into high frequency switching environments. Radial basis function neural networks have been combined with nonlinear sliding model control laws to reduce oscillations in DAB loops [8]. and XGBoost a Machine learning boosting algorithm has shown very good capability in predicting current oscillations in different kind of operating conditions [9]. For complex circuits with nonlinear configurations emerging algorithms used are Graph neural networks paired with multi layer perceptron to realize operational situation of multiple modulation types [10].

To manage sudden load changes, use of ai has increased to tune droop coefficients in DC microgrids to minimize active power deviations [11]. Also, deep reinforcement learning algorithms have been developed to track voltage fluctuations in vehicle to grid systems [12]. To eliminate ripples fuzzy based are used along with predictive models to get zero voltage switching across wide load margins [13]. state attention mechanisms with recurrent neural networks allow controllers to immediately balance control gains ensuring faster recovery during voltage spikes [14]. By using mathematics of DAB converter and integrating into neural network loss functions this will help in capturing transient inductor currents locus with high accuracy [15].

LSTM RNNs are deployed to optimize and make use of continuous data streaming sequences to make better decisions [16]. Advanced ripple reduction schemes are actively regulated by ML algorithms to smoothens out current fluctuations and preserve power quality [17]. Combining time series structural learning with isolated power converters balances sudden responses with excellent ripple reduction [18]. Predictive neural networks can be used to reduce total harmonic distortion improving power quality during extreme conditions [19]. To maintain charging current a effective modelling of SPS based closed loop operation for battery charging application [20]. To protect battery degradation during fast power, transfer a neuro fizzy controller strategy has been designed which helps in protecting the system from going into high temperature spikes [21].

1.1. Principle of Operation

DAB consists of two bridges each one consisting of four active controllable switches. One bridge act as inverter and other acts as rectifier. These two bridges are separated by a high frequency transformer which provides magnetic coupling and helps in energy

transfer between them. Primary bridge converts into DC voltage to high frequency square wave and secondary bridge produces another high frequency square wave. Power transfer depends on phase difference between two bridges. By shifting these waveforms relative to each other, a voltage drop is actively created across the series inductor, forcing energy to transfer.

1.2. Problem with normal closed loop configuration

Converters can either operate in open loop and closed loop configuration. In open loop mode there is no feedback from output to input so power transfer cannot be controlled after initialisation. It cannot respond for battery voltage fluctuations and load shifts. Closed loop solves this by utilising continuous feedback from output to adjust the phase difference between bridges.

However, implementing closed loop in high frequency conditions have some problems. If DAB converter switches on and off at high frequencies like 100Khz, the series inductor has high rate of change in current with time. Under such conditions if desired current suddenly changes high speed switching frequency causes high voltage ripples and high current ripples that will damage Battery.

1.3. Workflow and Project Motivation

The main motivation of this project is to improve the dynamic performance of the Dual Active Bridge (DAB) converter by reducing current and voltage ripples. The work begins with the design of a conventional dual-loop PI controller in Simulink. The input data generated from the PI controller is then used to train Artificial Neural Network and Recurrent Neural Network models. These trained AI models are combined with the existing PI controllers to assist the controller in improving performance. Finally, the performance of the conventional PI controller, PI with ANN, and PI with RNN-LSTM is compared under the same inputs.

Methodology:

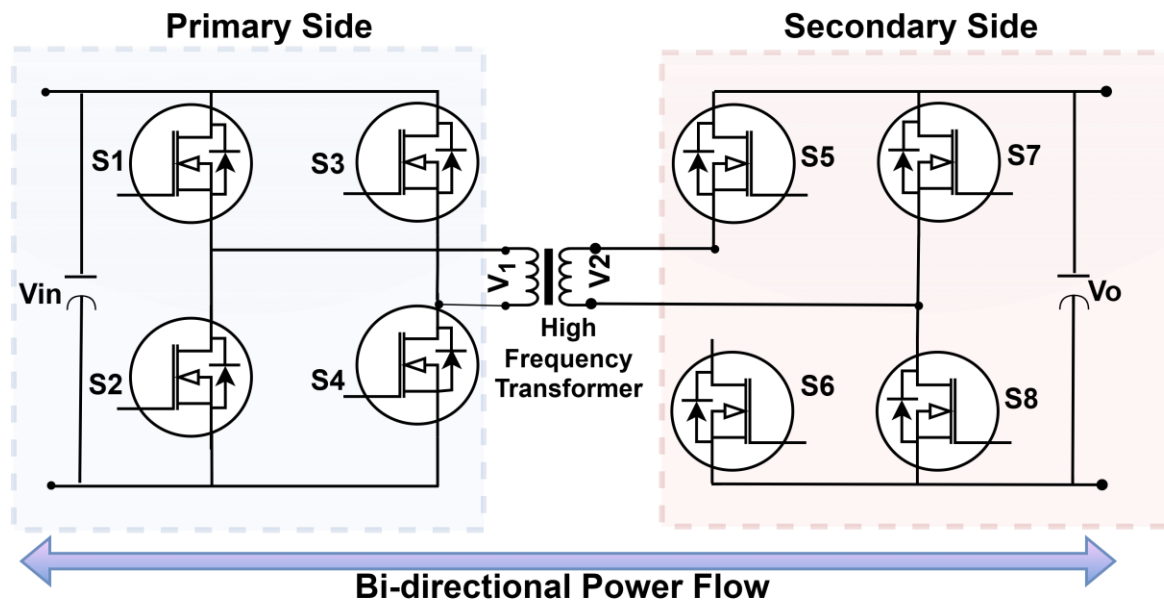
N =turns ratio (secondary by primary)

$$P = ((3.14 - s/3.14) * s * V_{in} * V_{out}) / (3.14 * n^2 * N * f)$$

$$L = (V_{in} * V_{out}) / (8 * N * f * P_{max})$$

Here P =average power, s is phase shift in radians

Parameters: 120V input, 48V battery payload, 20 microhenry inductor and 1320 microfarad capacitor



System runs at a switching frequency of 100 kHz inside a fixed discrete step size of 1 microsecond.

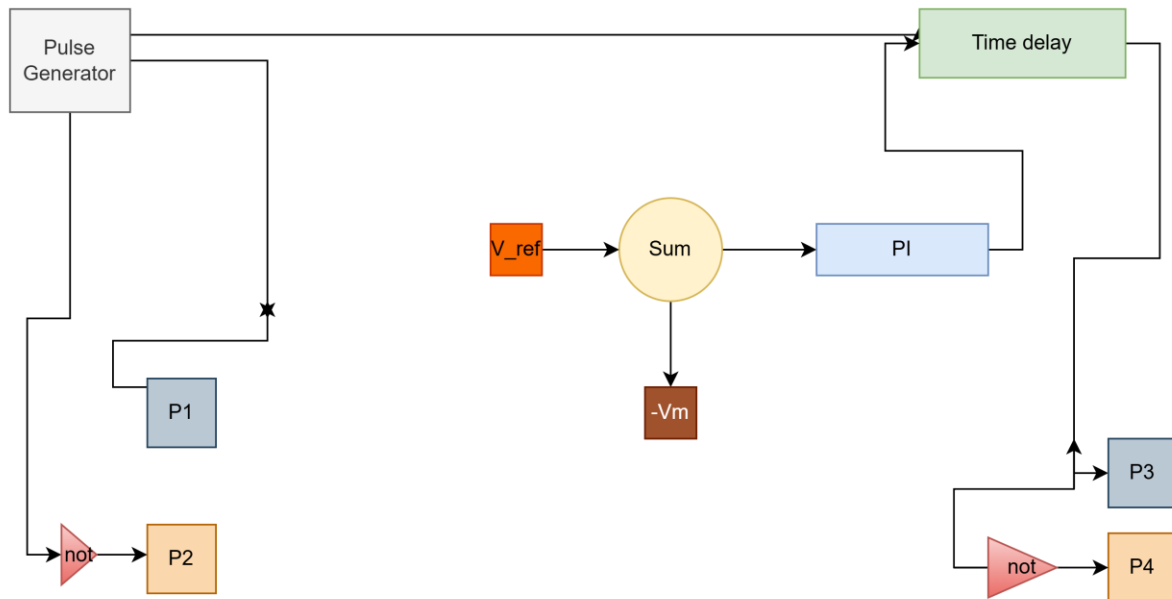
For this project, we set up a test to compare how different controllers handle current ripples and voltage ripples in a Dual Active Bridge converter. To make sure the test was completely comparable between different controllers, we kept the output voltage, output, current state of charge the same in all setups. Here battery is at output end, and we are charging a lithium battery.

The only part we changed from one test to the next was the control system itself. We tested three different setups in total: dual loop PI controller, dual loop PI with ANN support, and dual loop PI with RNN support. By just changing the controllers, we could see exactly how each design of the controller changed the performance of the system.

Single loop control

This is our simplest setup. Here, the output voltage feedback is directly compared against our target battery output voltage. The resulting error goes straight into a single PI controller block to generate the phase shift angle. Phase shift increases or decreases depends on output voltage. Here we have control over the voltage but no control over the current. While simple, this setup struggles because it cannot see or limit the rapid current spikes flowing into the battery load.

The control circuit is shown below

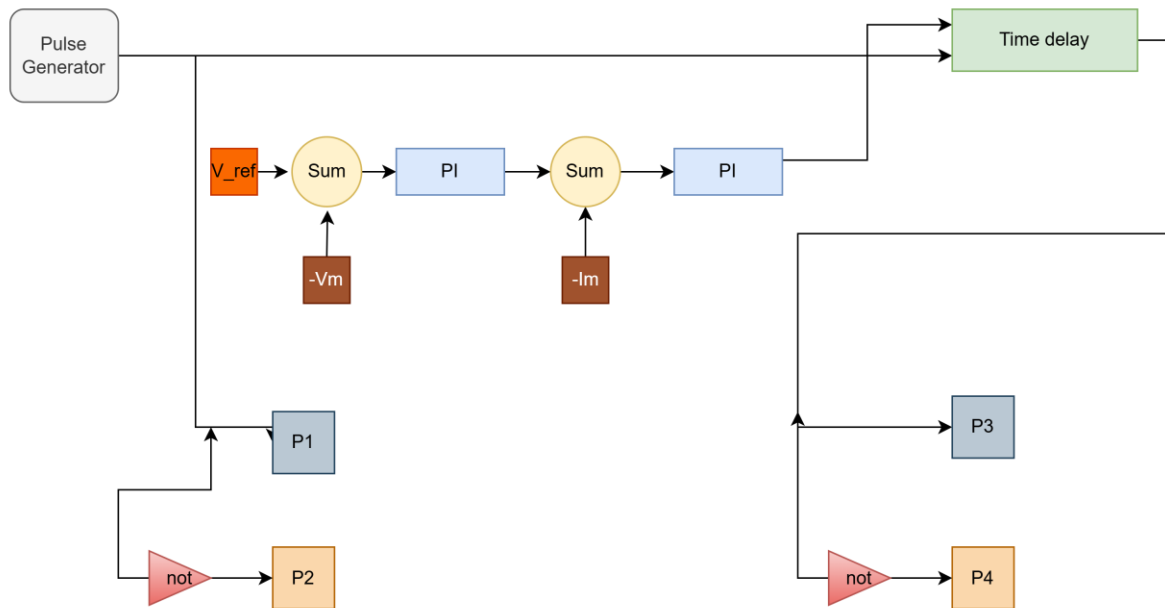


Case1: Dual loop control

To control both current and voltage, we have designed the system by nesting two separate control loops together in a cascaded structure. The outer loop monitors the output voltage error and uses a PI block to calculate a dynamic current reference. This calculated reference is then handed down to an inner loop, which checks the battery current error continuously through a second PI block to output the final phase shift angle.

By adding this second loop we are controlling both current and voltage at the same time. We added extra control of safety compared to single loop control. We need steady current and steady voltage charging. In this configuration lot of ripples in current and voltage are observed but between two values.

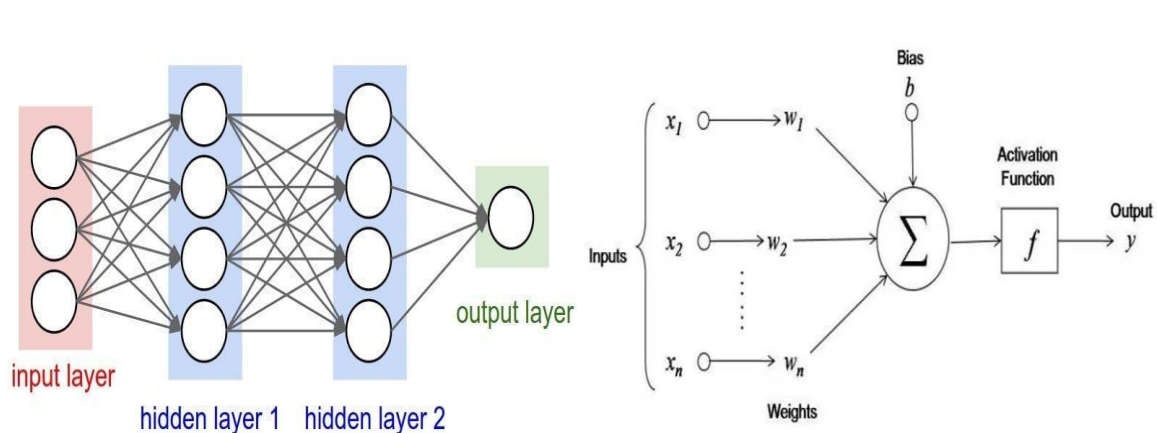
The control circuit is shown below



Case 2: Dual loop with Artificial neural network

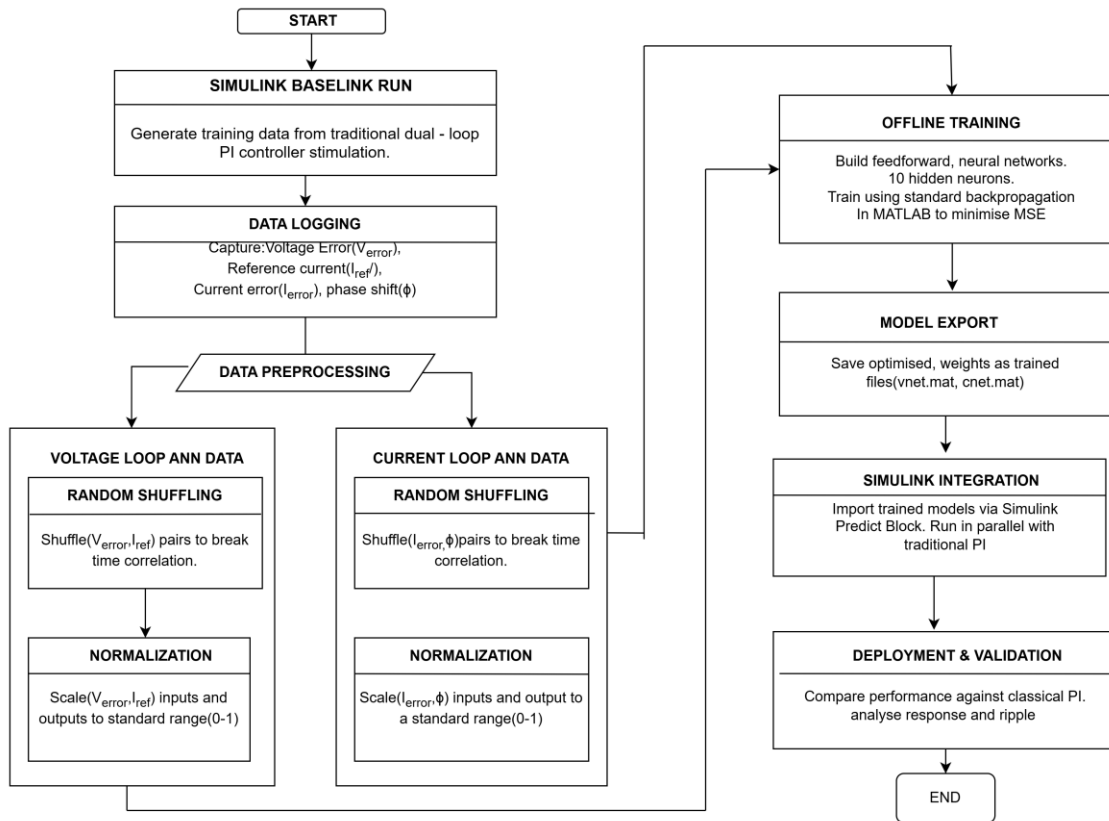
In this setup we used ANN as support for both PI controllers. An artificial neural network is a deep learning model inspired from human brain. It consists of interconnected neurons organized into an input layer, hidden layers and a output layer. The network learns patterns from input data by correcting connection weights during training, this will help in predicting new data accurately. We have used 10 hidden neurons in our setup.

To implement this configuration, we split neural network blocks into two different types, a voltage assister neural network for outer loop and current assister neural network for inner loop. Training data was generated from normal dual loop configuration from above. Four parameters were collected; they are voltage error and 1st PI controller output (reference current) and current error and second PI controller output (phase shift). Random sampling was done because data collected is huge.



Inputs for voltage loop neural network are error voltage and reference current. Inputs for current loop neural network are error current and phase shift. Before feeding data into network, we pass it through data normalisation stage. If we train data on a unnormalized

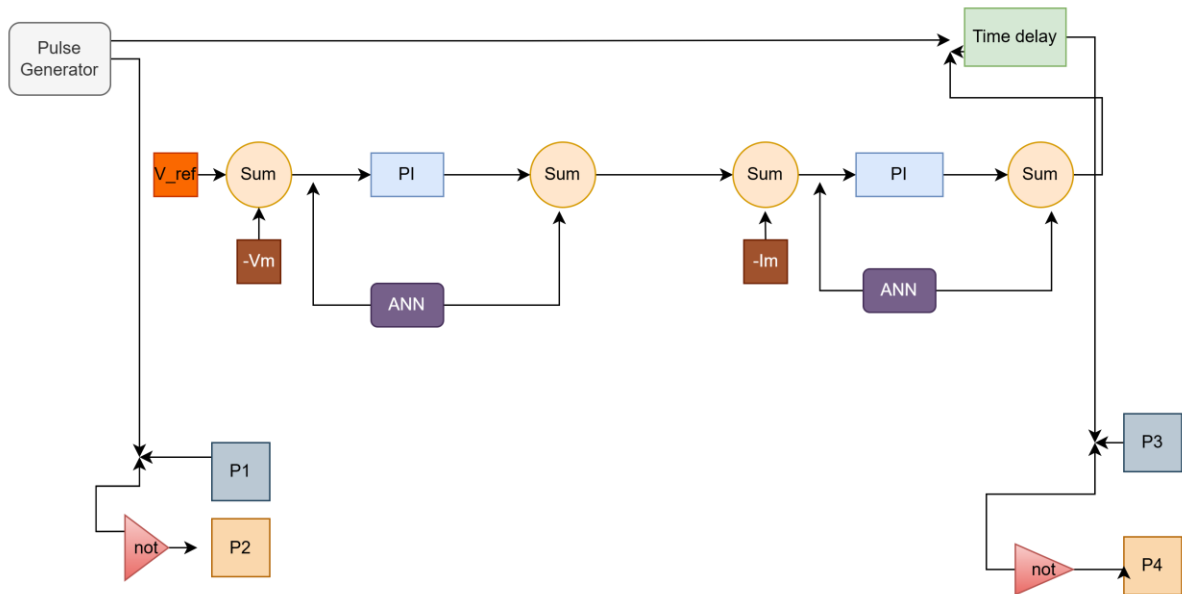
data, the larger numerical values will dominate weight adjustments, so the model can't learn properly which can lead to poor performance. We used epoch of 200 which means the model will see each input point 200 times. The flowchart shows the workflow for ANN based controller is shown below.



So, Normalisation eliminates bias by allowing training algorithm to treat all features equally. Later we use normalised and denormalized blocks in simulation to get correct results. After scaling data, we constructed feedforward neural network structure consisting of a 10-neuron hidden layer. This network was trained using backpropagation algorithm and final sampled data was stored.

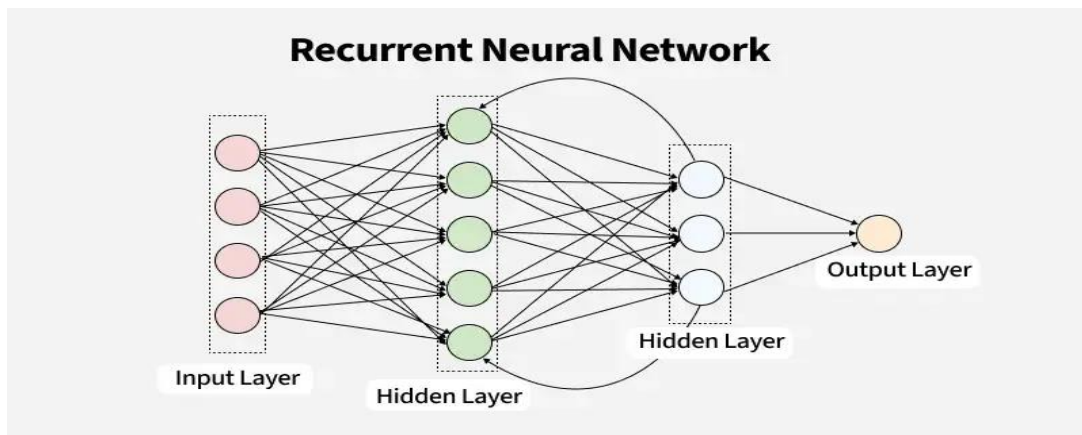
Once both networks were fully trained, we uploaded the saved data files back into simulation using Simulink predict block. This allowed both networks to run in parallel with our dual PI controller. These two networks helped both PI controllers to make faster and better decisions. Instead of depending purely on fixed, linear PI formulas that struggle with non-linear changes like sudden increase in current or voltage, the network instantly evaluates voltage and current errors and inject a immediate phase angle correction.

The control circuit for ANN based controller is shown below.

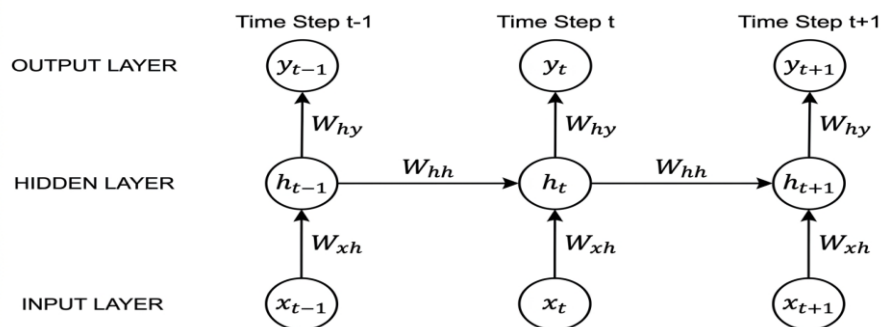


Case 3: Dual loop with Recurrent neural network

In the last setup, we used a Recurrent Neural Network (RNN), a Long Short-Term Memory (LSTM) network, as a support for both PI controllers. ANN is a stateless feedforward model; an RNN is a deep learning model which remembers some memory from its past. It consists of interconnected memory cells that include hidden states and cell states.

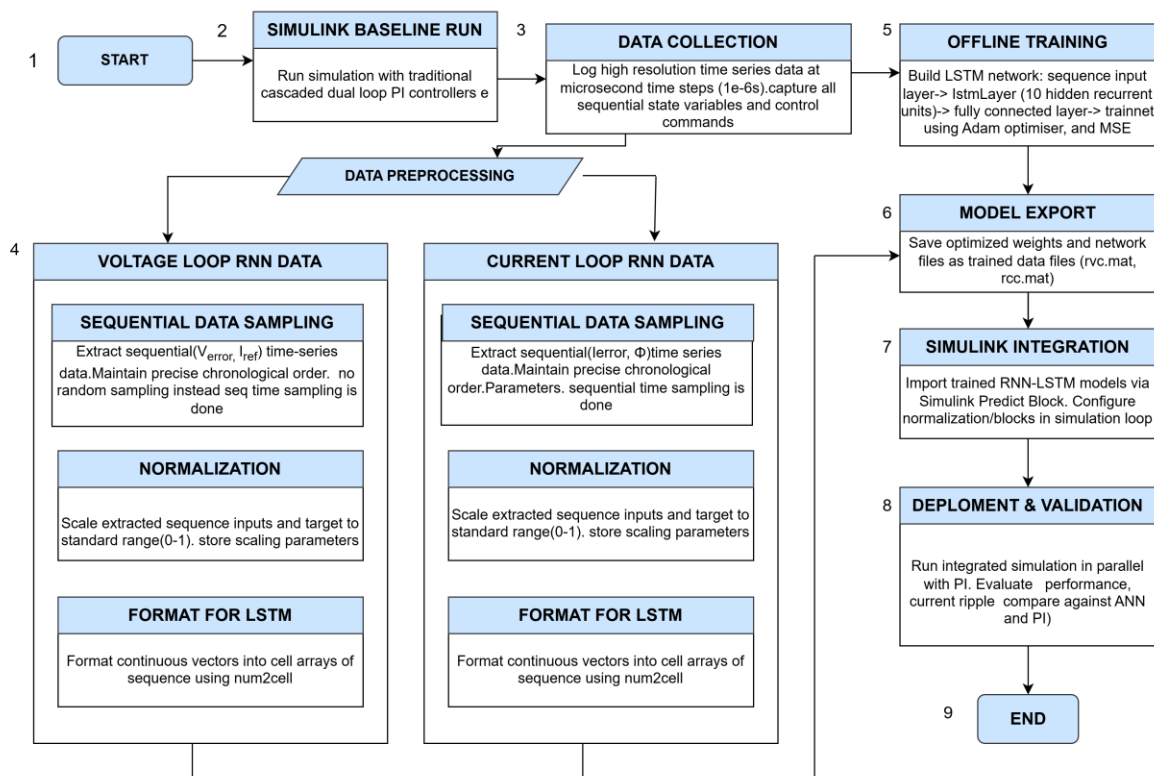


UNROLLED RECURRENT NEURAL NETWORK (RNN) ARCHITECTURE AND SHARED WEIGHTS



These loops allow the network to get information from previous time steps, meaning it can learn patterns from input data over time by adjusting connection weights during training.

To implement this configuration, we divided the recurrent network architecture into the same two-stage one is outer voltage loop and other is inner current loop layout same as ANN setup. The training data was generated from the same dual-loop configuration. Sequential time series sampling was done to the generated training data here unlike ANN where we did random sampling due to its stateless nature. We stored continuous data to feed memory to RNN. For the outer voltage network and inner current network, this data specifically collected $N = 20000$ sequential samples of the voltage error and the first PI controller output (VPI). We used epoch of 150 which means the model will see each input point 150 times. The flowchart shows workflow of RNN based controller as shown below.



To ensure consistency and eliminate numerical bias, these vectors were processed through the exact same normalization bounds before training.

For the training, instead of the feedforward backpropagation used in the ANN, we trained this network structure using the Adam optimizer to minimize the mean square error over 150 epochs and seen the performance on a training plot over time. This training period continuously updated recurrent weights and finally we will get optimal weights.

Once both networks were fully trained, we stored the resulting data files(.mat). We then uploaded these saved network files back into the simulation using the Simulink Predict

block, allowing both recurrent blocks to execute in parallel with our traditional PI controller.

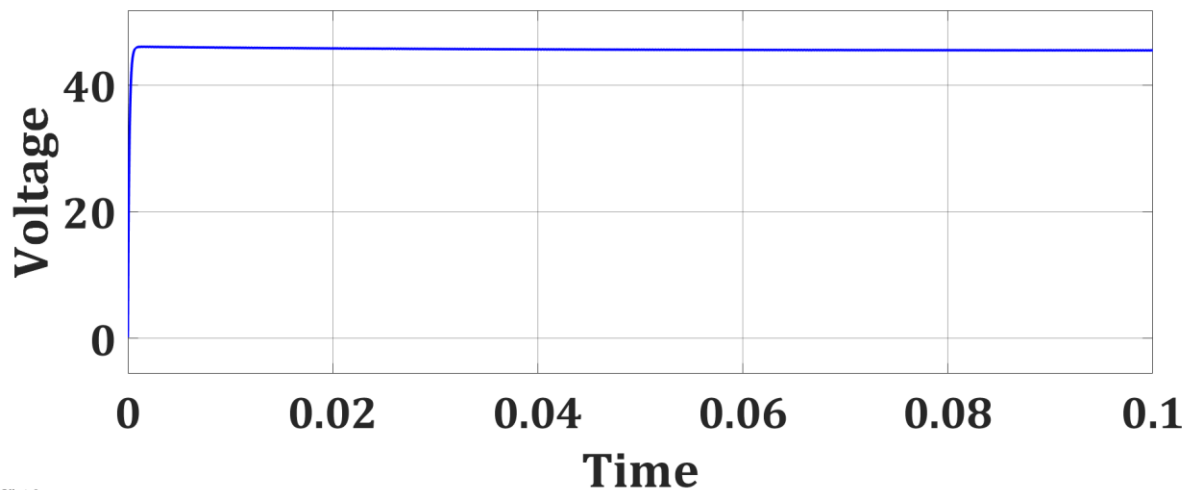
While these networks also helped PI controllers to make faster and best decisions at time of instability, the critical difference lies in their internal recurrent layer. Instead of looking at a single instant like the ANN, the hidden layer with its 10 recurrent units tracks the progression from one step to the next over the execution loop. This internal tracking allows the network to give a highly precise and reliable phase angle correction.

Results:

The simulation results for various type of controllers are shown below

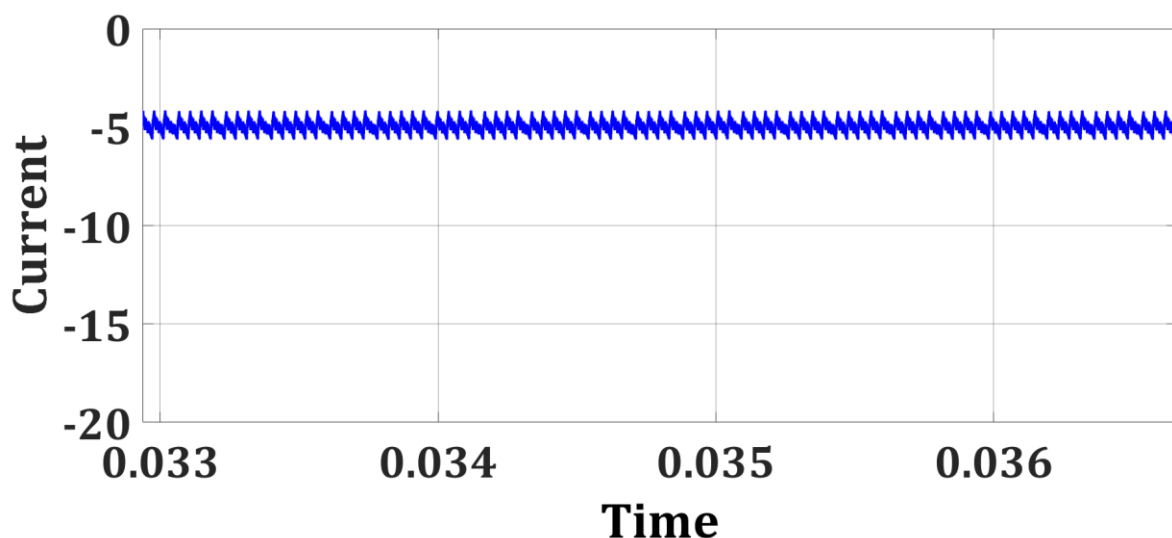
For Dual loop

1. Battery Voltage vs time

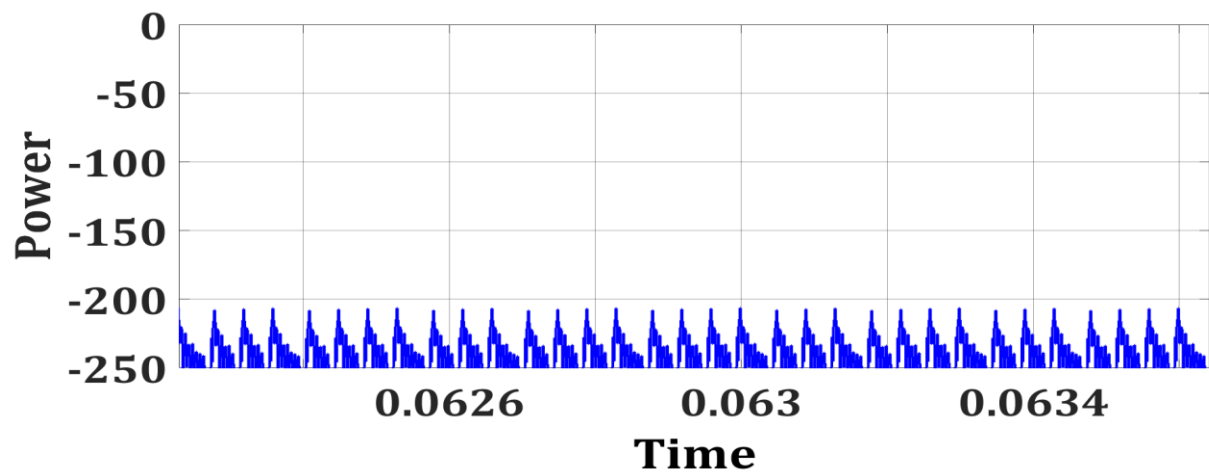


Offset=0

2. Battery current vs time

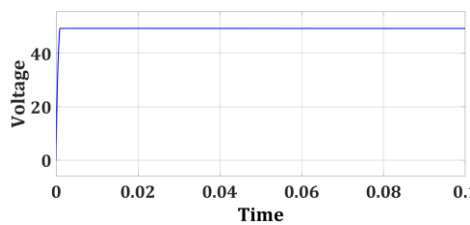


3. Power vs time

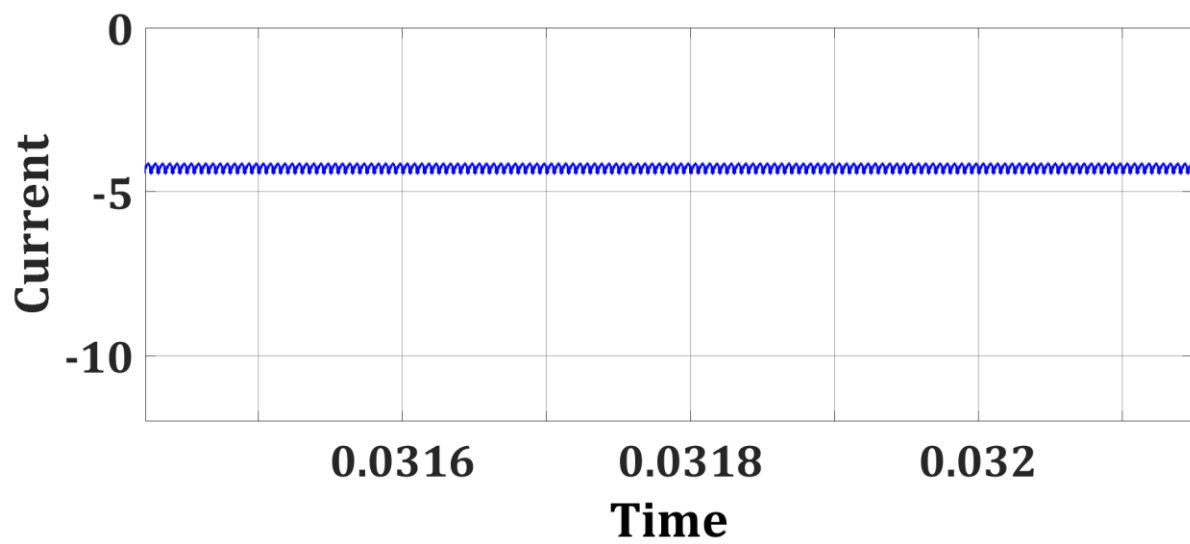


For dual loop with ANN

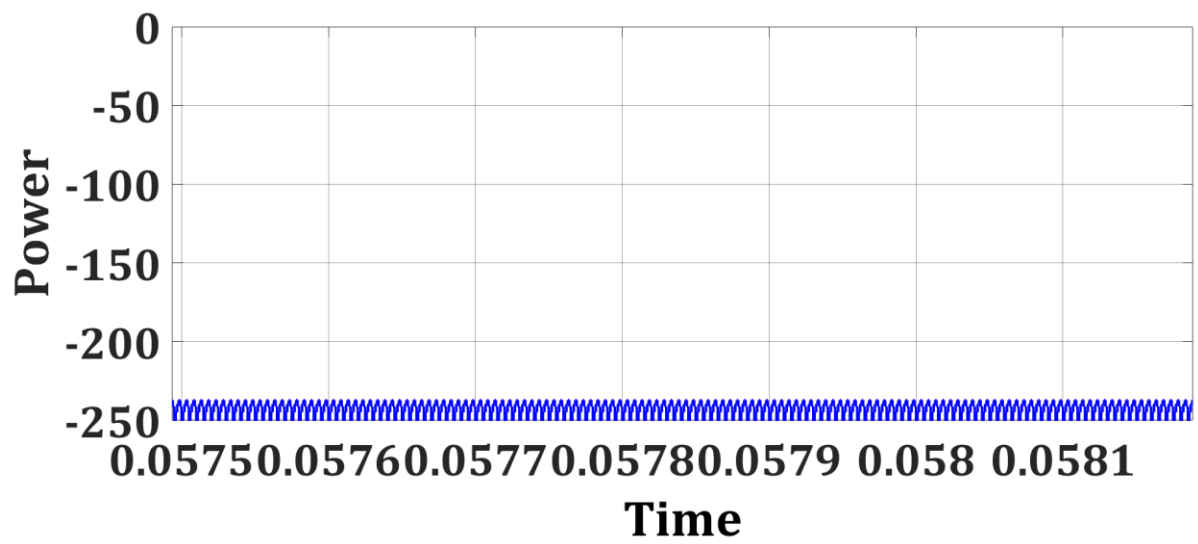
1. Battery Voltage vs time



2. Battery current vs time

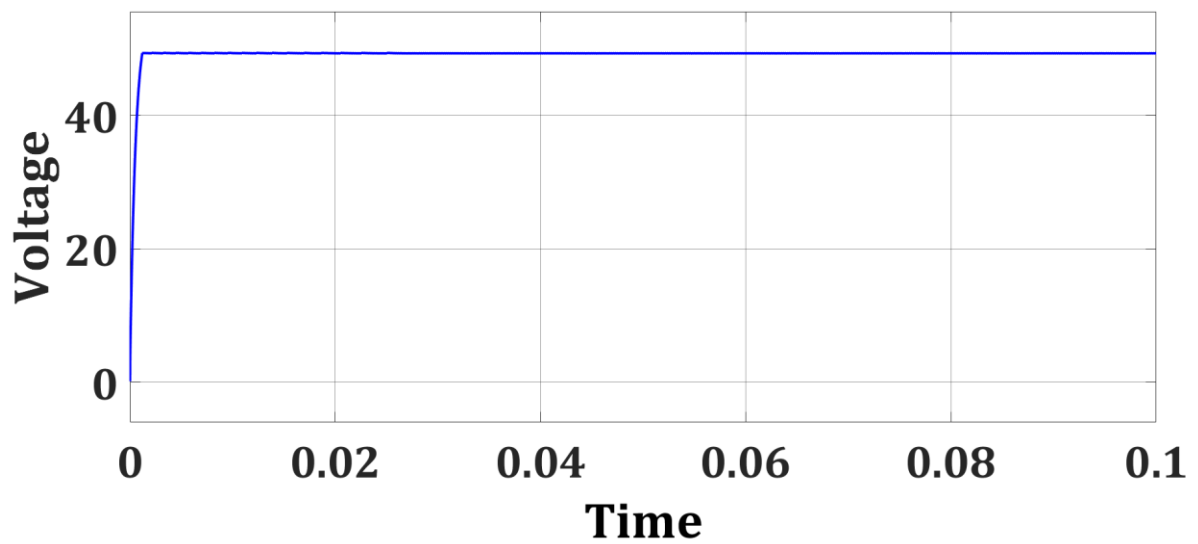


3. Power vs time

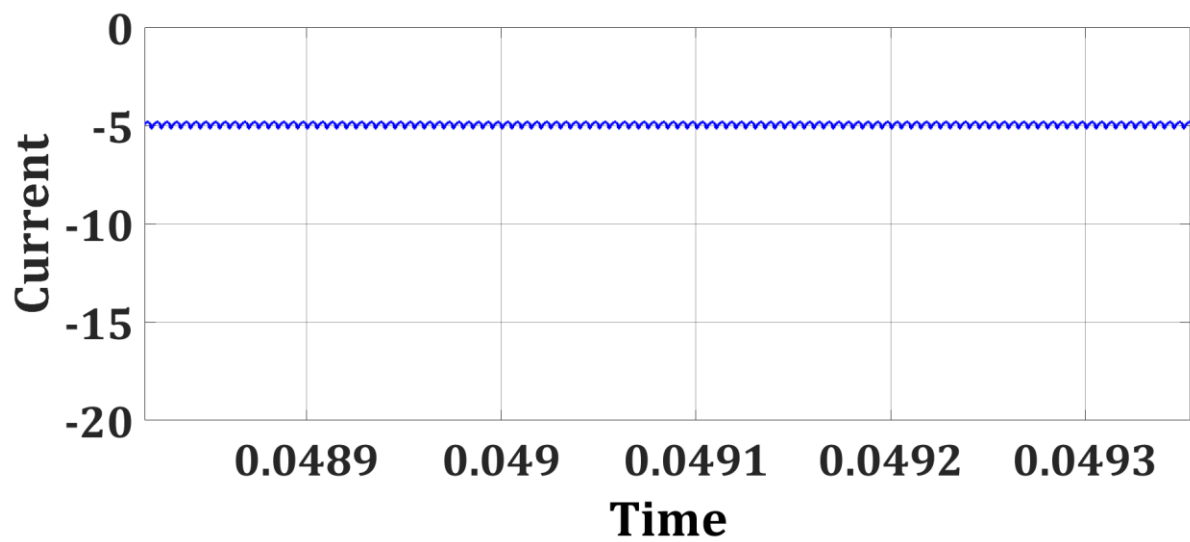


For dual loop with RNN

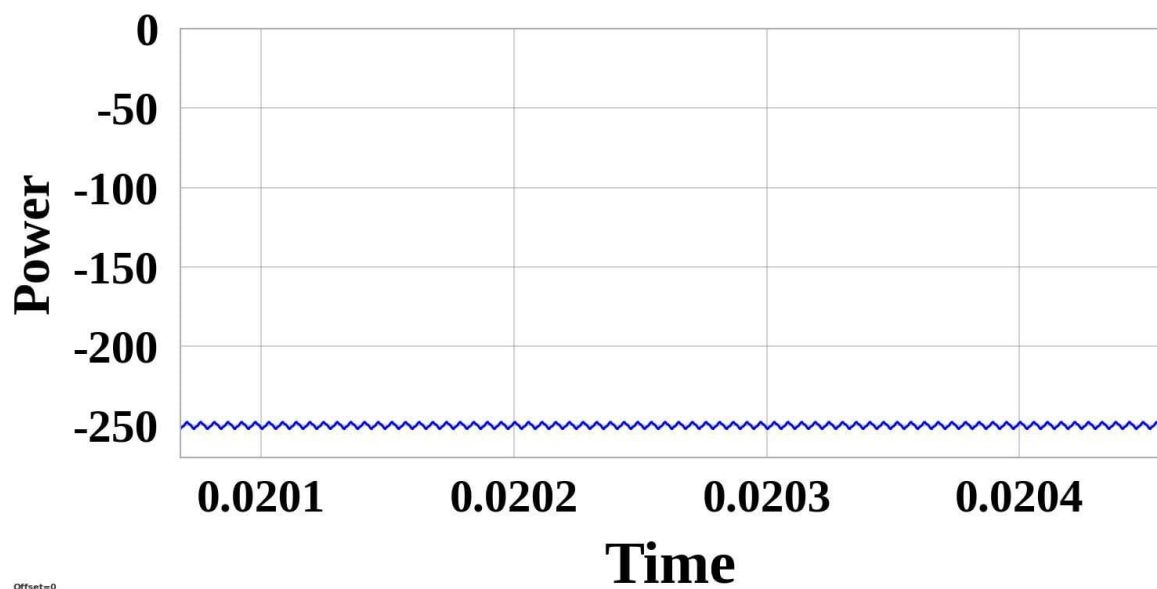
1. Battery Voltage vs time



2.current vs time



3.Power vs time



Offset=0

Comparison of results:

PARAMETERS	PERFORMANCE METRIC	CASCADED DUAL-LOOP PI	PARALLEL ANN- ASSISTED	RECURRENT RNN- LSTM
VOLTAGE (V) Figure:1,4,7	SETTLING TIME	Slow 2.50 ms	Fast 1.20 ms	Fastest 0.95 ms
	OVERSHOOT	Highest 0.85%	Low 0.21%	Lowest 0.05%

	RESPONSE CHARACTERISTIC	Linear response with initial delay.	Fast rise time via non-linear gain	Rose and settled faster than ANN model
CURRENT (I) Figure:2,5,8	RIPPLE PROFILE	High oscillations of nearly 2.4 A	Fixed ripple =1.85 A	Minimized ripple = 0.35 A
	WAVEFORM QUALITY	Regular triangular wave	High ripple noisy waveform	Clean waveform with low noise
	SWITCHING BEHAVIOR	Standard linear error bounds from fixed switching	Stateless control results in high frequency chatter	Internal memory tracks trends and dampens high-frequency noise.
POWER (P) Figure:3,6,9	STABILITY BOUNDS	Cyclic oscillations	Chaotic spikes	Constant steady state
	NOISE / JITTER	Moderate periodic noise	Severe jitter $\pm$ 52 W	Negligible noise $\pm$ 4.5 W
	TARGET TRACKING ~250 W	Oscillates continuously between ~210 W and ~250 W	Inconsistent tracking due to switching jitter	Maintain stable active power at target

Conclusion:

The standard dual-loop PI controller gave us a good control of current and voltage. But we need constant current and constant voltage charging which took lot of time to give. Initially there were lot of oscillations at start for both voltage and current so settling time is more here. Because it relies on static formulas, the controller takes too little, long to adjust compared to other controllers, leading to temporary voltage drops and high-frequency current ripples before settling down.

The ANN helper loop: Adding feedforward ANN solved slow response of classical PI setup. It instantly adjusted phase shift angle. As this network is stateless and has no memory it has significant ripples in the current, but lesser than Dual PI.

The RNN helper loop: RNN gave the overall best performance, It has used 10 hidden neurons to track history of data from previous steps. It took decisions from memory it stored. Current waveform is smoother than other two.

Future scope:

Future scope includes testing of the designed AI based controllers in real-time using OPAL-RT and its hardware implementation for battery charging.

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